Unit- 4 Renewable Energy

Energy Basics — Conventional Energy

1. Energy

Definition: Energy is the capacity to do work.

Energy is required for:

- transportation
- electricity generation
- industrial processes
- domestic activities

Example: electricity, heat, fuel energy.

2. Conventional Energy

Definition : Conventional energy refers to traditional energy sources that have been used for a long time to meet human energy needs.

These sources are mainly **fossil fuels** and are usually **non-renewable**.

They were the **primary sources of energy during industrial development**.

Main Characteristics of Conventional Energy

Characteristic	Explanation
Traditional sources	Used for many decades
Mostly non-renewable	Limited availability
High energy output	Large-scale electricity generation
Environmental impact	Causes pollution and greenhouse gases

Examples of Conventional Energy

1. Coal : Coal is a **fossil fuel formed from plant remains buried underground for millions of years**.

Uses

- thermal power plants
- steel production
- cement industry

Advantages

- abundant in many countries
- high energy content

Disadvantages

- air pollution
- carbon dioxide emissions

2. Petroleum (Crude Oil)

Petroleum is a **liquid fossil fuel found underground and refined into many products**.

Common Products

- petrol
- diesel
- kerosene
- lubricants

Uses

- transportation fuel
- petrochemical industries
- electricity generation

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Problems

- air pollution
- limited reserves

3 Natural Gas

Natural gas is a **gaseous fossil fuel mainly composed of methane**.

Uses

- cooking fuel (LPG / PNG)
- electricity generation
- fertilizer production

Advantages

- cleaner than coal and petroleum
- high efficiency

Disadvantages

- still produces greenhouse gases
- limited supply

Renewable Energy

1 Definition : Renewable energy is energy obtained from natural sources that are **continuously replenished and cannot be exhausted by human use**.

These energy sources are naturally renewed through **environmental processes**.

Examples include:

- solar energy
- wind energy
- hydro energy
- biomass energy
- geothermal energy

2. **Meaning of Renewable Resources**

Renewable resources are natural resources that can be **replaced or regenerated naturally within a short period of time**.

They are **sustainable** because their supply is continuously renewed by nature.

Examples of Renewable Resources

Resource	Source
Solar energy	Sunlight
Wind energy	Moving air
Hydropower	Flowing water
Biomass	Organic matter
Geothermal energy	Heat from Earth

Conventional Energy vs Renewable Energy

Feature	Conventional Energy	Renewable Energy
Definition	Traditional energy sources used for a long time	Energy obtained from naturally replenished resources
Availability	Limited and exhaustible	Abundant and continuously replenished
Examples	Coal, petroleum, natural gas	Solar, wind, hydro, biomass
Pollution	Produces high pollution and greenhouse gases	Very low or no pollution
Sustainability	Not sustainable for long-term use	Sustainable and environmentally friendly

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Cost	High environmental cost	Lower environmental impact

Need for Renewable Energy

Renewable energy is important to meet the **growing energy demand while protecting the environment and natural resources**.

1. Energy Crisis

Meaning : An energy crisis occurs when the demand for energy exceeds the available supply of energy resources.

Most conventional energy sources such as **coal, petroleum, and natural gas are limited and non-renewable**. Rapid industrialization and population growth increase energy consumption, leading to shortages.

Role of Renewable Energy

Renewable energy sources such as **solar, wind, and hydro energy** provide alternative energy and help reduce dependence on fossil fuels.

2. Pollution Reduction

Conventional energy sources produce large amounts of **air pollution and greenhouse gases**.

Examples:

- carbon dioxide (CO_2)
- sulfur dioxide (SO_2)
- nitrogen oxides (NO_x)

These pollutants cause:

- global warming
- climate change
- respiratory diseases

Benefit of Renewable Energy

Renewable energy sources produce **little or no pollution**, helping to protect the environment and improve air quality.

3. Sustainability

Meaning : Sustainability refers to the ability to meet present energy needs without compromising the ability of future generations to meet their own needs.

Fossil fuels are **exhaustible**, meaning they will eventually run out.

Renewable energy sources are **naturally replenished**, making them suitable for long-term energy supply.

Summary Table

Reason	Explanation
Energy crisis	Renewable energy helps overcome shortage of fossil fuels
Pollution reduction	Produces less pollution compared to conventional energy
Sustainability	Provides long-term and continuous energy supply

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Solar Energy

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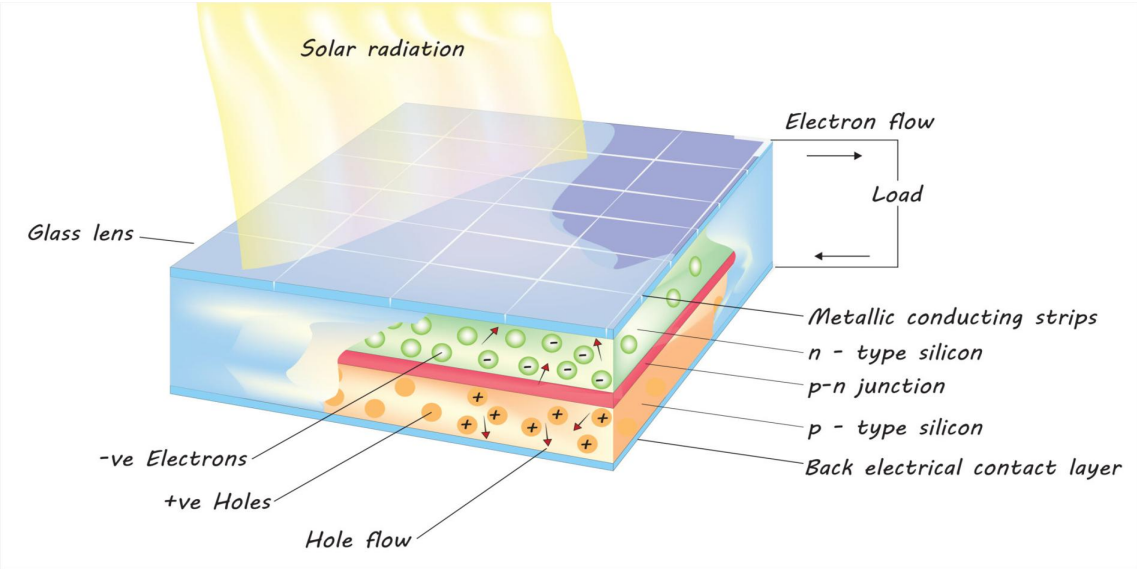
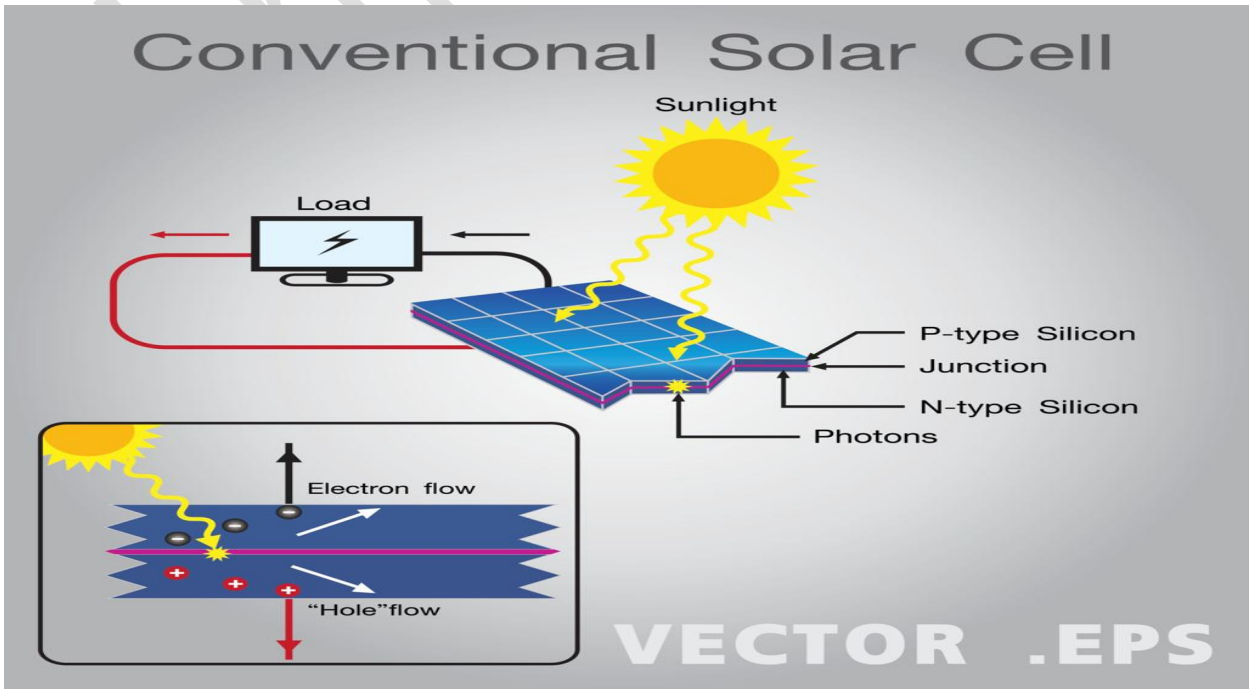


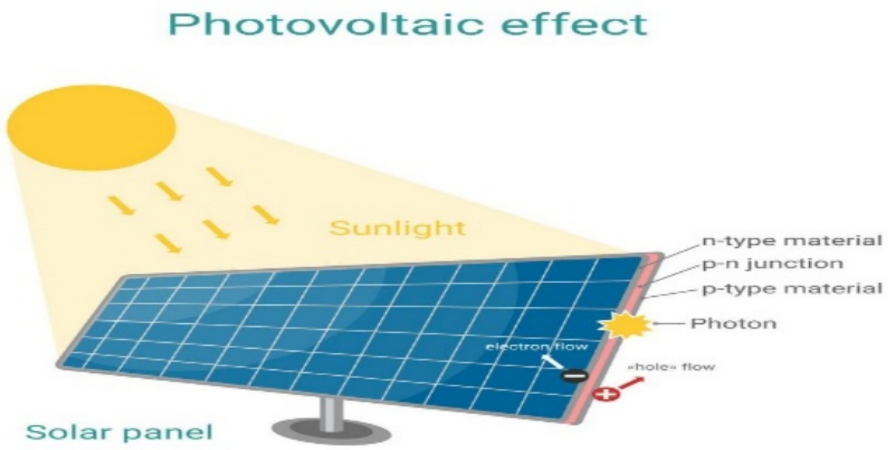
Definition : Solar energy is the energy obtained from the **sun's radiation** and converted into useful forms such as **electricity or heat**.
Solar energy is a **renewable and clean source of energy** because sunlight is continuously available and does not produce pollution.

Characteristics of Solar Energy

Feature	Explanation
Renewable	Sunlight is continuously available
Clean energy	Produces no air pollution
Abundant	Available in most parts of the world
Sustainable	Can be used for long-term energy supply

Solar Cell
Definition: A solar cell is a device that converts **sunlight directly into electrical energy** using the **photovoltaic effect**.
Solar cells are the basic units used in **solar panels**.





Principle of Solar Energy (Photovoltaic Effect)
Solar cells work on the **photovoltaic effect**, in which light energy is converted into electrical energy. When **sunlight falls on a semiconductor material** (usually silicon), electrons gain energy and start moving, producing **electric current**.

Working of a Solar Cell
The working process occurs in several steps:
Step 1 — Sunlight Absorption
Sunlight falls on the **solar panel** containing photovoltaic cells.

Step 2 — Electron Excitation
Photons from sunlight transfer energy to electrons in the **semiconductor material**.
Electrons become energized and move freely.

Step 3 — Electric Field Action
The **p-n junction** inside the solar cell creates an electric field.
This field forces electrons to move in a particular direction.

Step 4 — Electric Current Generation
The movement of electrons through an external circuit produces **direct current (DC) electricity**.

Step 5 — Power Utilization
The electricity generated can be used for:
- lighting
- charging batteries
- running electrical appliances

Components of a Solar Cell

Component	Function
Glass cover	Protects the cell
Anti-reflective coating	Increases sunlight absorption
Semiconductor layer	Generates electric charge

Metal contacts	Collect and transfer electricity

Applications of Solar Energy

Solar energy is used in many areas:

- solar panels for electricity
- solar water heaters
- solar street lights
- solar cooking systems
- satellites and space stations

Advantages of Solar Energy

Advantage	Explanation
Renewable	Unlimited energy source
Environment friendly	No pollution
Low operating cost	Sunlight is free
Widely available	Accessible in many regions

Limitations of Solar Energy

Limitation	Explanation
Weather dependent	Requires sunlight
High installation cost	Solar panels are expensive
Energy storage required	Batteries needed for night use

Solar Energy — Advantages, Limitations, Applications

Advantages

Advantage	Explanation
Clean energy	Solar energy does not produce harmful gases or air pollution.
Abundant	Sunlight is widely available and renewable.
Low operating cost	After installation, maintenance and running costs are very low.

Limitations

Limitation	Explanation
High installation cost	Initial cost of solar panels and equipment is high.
Weather dependency	Energy production depends on sunlight and weather conditions.

Applications

Application	Description
Solar panels	Used to generate electricity from sunlight.
Solar water heaters	Used for heating water using solar energy.
Solar farms	Large-scale solar power plants that produce electricity for cities and industries.

Top of Form

Bottom of Form

Wind Energy

Definition : Wind energy is the energy obtained from the **movement of air (wind)** and converted into useful forms such as electricity using wind turbines.

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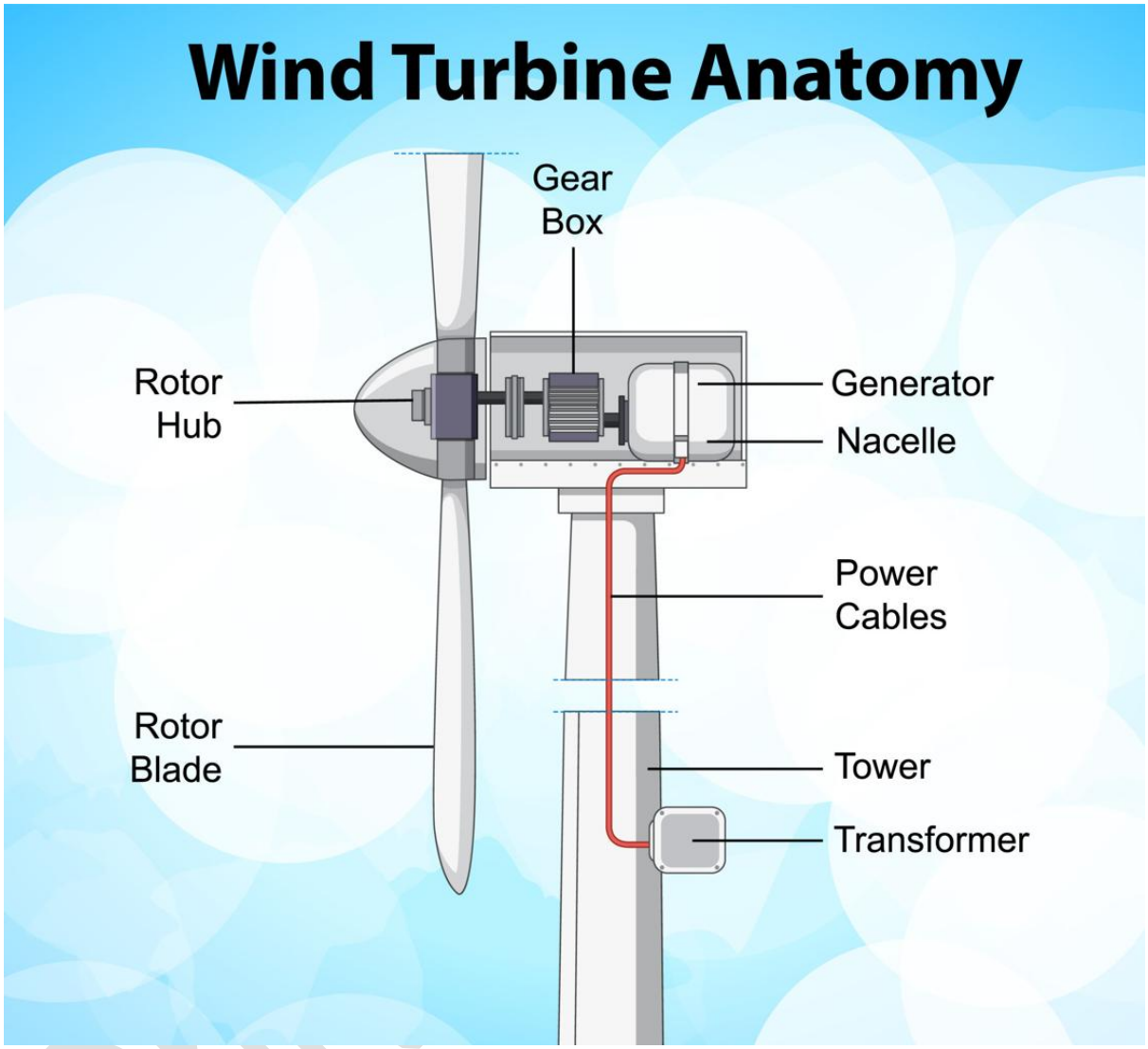
Wind energy is a **renewable and clean source of energy** because wind is naturally produced by atmospheric processes and does not get exhausted.

Principle of Wind Energy



Wind energy works on the principle of **conversion of kinetic energy of moving air into mechanical energy and then into electrical energy**.
Steps involved in the principle:

1. Wind flows due to **difference in air pressure caused by uneven heating of the Earth by the sun**.
2. Moving air contains **kinetic energy**.
3. When wind strikes the blades of a turbine, it causes them to **rotate**.
4. The rotating blades turn a **shaft connected to a generator**.
5. The generator converts mechanical energy into **electrical energy**.

Wind Turbines : A wind turbine is a machine that converts the **kinetic energy of wind into mechanical energy and then into electrical energy**.
Wind turbines are usually installed in **wind farms** where wind speed is high and consistent.

Main Components of a Wind Turbine

Component	Function
Blades	Capture the kinetic energy of wind
Rotor	Rotates when blades move
Shaft	Transfers mechanical energy
Generator	Converts mechanical energy into electricity
Tower	Supports turbine at high altitude
Nacelle	Contains gearbox and generator

Working of Wind Turbines

The working process occurs in several stages:

Step 1 — Wind Flow
Wind moves across the turbine blades.

Step 2 — Blade Rotation
The wind force causes the **blades to rotate**, turning the rotor.

Step 3 — Shaft Rotation
The rotor is connected to a **shaft**, which rotates due to blade movement.

Step 4 — Generator Operation
The rotating shaft spins a **generator**, converting mechanical energy into electricity.

Step 5 — Electricity Transmission
Generated electricity is transferred to **power grids for use in homes and industries**.

Applications of Wind Energy
Wind energy is widely used for:

- electricity generation
- wind farms
- water pumping
- agricultural irrigation
- remote power supply systems

Advantages of Wind Energy

Advantage	Explanation
Renewable	Wind is naturally replenished
Clean energy	Produces no air pollution
Low operating cost	No fuel cost
Sustainable	Can be used long-term

Limitations of Wind Energy

Limitation	Explanation
Weather dependent	Requires sufficient wind speed
Noise pollution	Turbine operation produces sound
Land requirement	Wind farms need large areas

Hydropower (Hydroelectric Energy)
Definition : Hydropower is the energy produced by using the **force of flowing or falling water to generate electricity**.
It is a **renewable source of energy** because the water cycle continuously replenishes water resources.
Hydropower is widely used in **hydroelectric power plants**.

Principle of Hydropower Plant
Hydropower plants work on the principle of **conversion of potential energy of stored water into mechanical energy and then into electrical energy**.
Steps of the principle:

1. Water is stored in a **dam at a higher elevation**.
2. Stored water has **potential energy**.
3. When water flows down through pipes (penstocks), its potential energy converts into **kinetic energy**.
4. Flowing water strikes the **turbine blades**, causing them to rotate.
5. The rotating turbine drives a **generator**.
6. The generator converts mechanical energy into **electrical energy**.

Hydropower Plant
A **hydropower plant** is a facility that generates electricity using the **energy of flowing water**.

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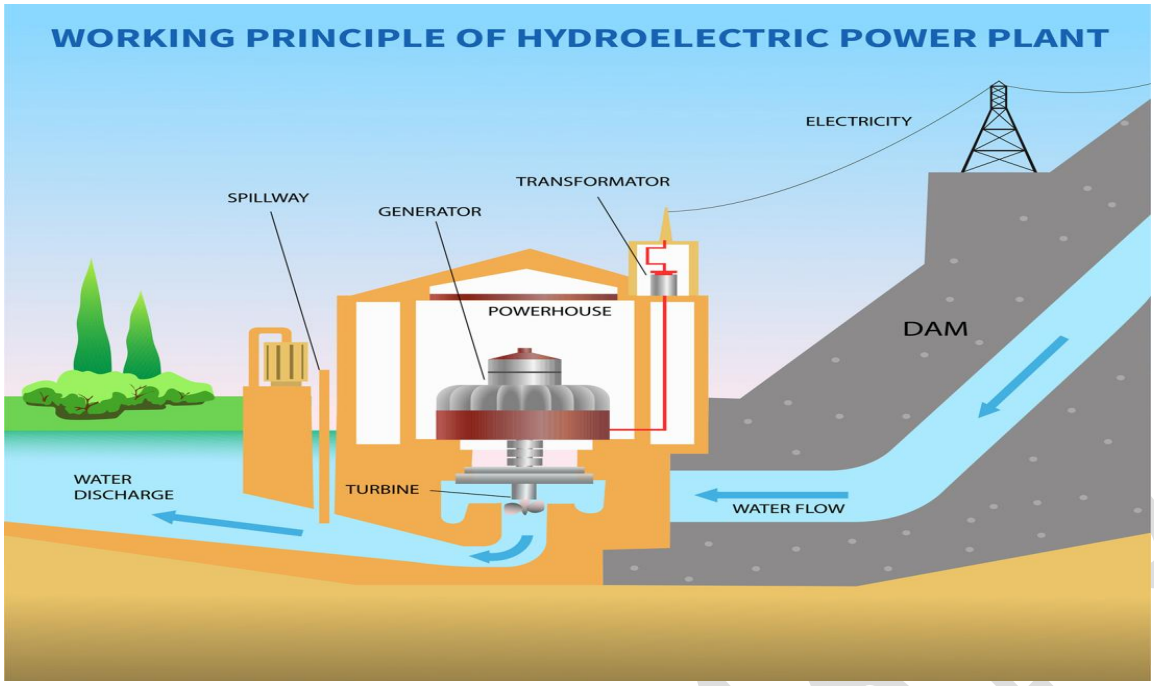

Main Components of a Hydropower Plant

Component	Function
Dam	Stores water at high level
Reservoir	Large water storage behind dam
Penstock	Pipe carrying water to turbine
Turbine	Rotated by flowing water
Generator	Converts mechanical energy into electricity
Transmission lines	Carry electricity to consumers

Working of a Hydropower Plant

Step 1 — Water Storage

Water is stored in a **reservoir behind the dam**.

Step 2 — Water Flow

Water is released from the reservoir through **penstocks**.

Step 3 — Turbine Rotation

Flowing water strikes the **turbine blades**, causing them to rotate.

Step 4 — Generator Operation

The turbine rotates a **generator**, producing electricity.

Step 5 — Power Distribution

Electricity is transmitted through **power lines** to homes and industries.

Energy Conversion in Hydropower

Energy Type	Conversion
Potential energy	Stored water in dam

Kinetic energy	Flowing water
Mechanical energy	Rotating turbine
Electrical energy	Generator output

Applications of Hydropower

Hydropower is used for:

- large-scale electricity generation
- irrigation projects
- flood control
- water supply systems

Advantages of Hydropower

Advantage	Explanation
Renewable	Water cycle continuously replenishes supply
Clean energy	Produces very low pollution
Reliable	Stable electricity generation
Low operating cost	No fuel required

Limitations of Hydropower

Limitation	Explanation
High construction cost	Dam construction is expensive
Environmental impact	Affects aquatic ecosystems

Hydropower — Advantages, Limitations, Examples

Advantages

Advantage	Explanation
Reliable	Hydropower plants can generate electricity continuously when sufficient water is available.
Renewable	Water is replenished through the natural **water cycle**, making hydropower a renewable energy source.

Limitations

Limitation	Explanation
Ecosystem disruption	Construction of dams can disturb aquatic life and natural river ecosystems.
High construction cost	Building dams and hydropower plants requires large initial investment and infrastructure.

Examples

Example	Description
Hydroelectric dams	Large dams built across rivers to generate electricity using flowing water.

Biomass Energy

Definition : Biomass energy is the energy obtained from **organic materials such as plants, agricultural waste, animal waste, and other biological matter**.

These materials are collectively called **biomass**, and they can be used as fuel to produce **heat, electricity, or biofuels**.

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Biomass energy is considered **renewable** because the organic materials used can be **regenerated through natural processes**.

Meaning of Biomass
Biomass refers to **organic matter derived from living or recently living organisms**.
Examples of biomass materials:

- wood and plant residues
- agricultural waste (crop residues)
- animal manure
- food waste
- forest residues

These materials contain **stored chemical energy from the sun**, which plants absorb during **photosynthesis**.

Principle of Biomass Energy
Biomass energy works on the principle of **conversion of chemical energy stored in organic matter into heat or electricity**.
Plants absorb **solar energy through photosynthesis**, storing it in the form of chemical energy. When biomass is burned or processed, this energy is released and used for energy production.

Conversion Methods of Biomass Energy
Biomass energy can be converted into usable energy through different methods.

Method	Explanation
Direct combustion	Biomass is burned to produce heat and electricity
Anaerobic digestion	Organic waste is decomposed by microorganisms to produce biogas
Fermentation	Biomass is converted into biofuels such as ethanol
Gasification	Biomass is converted into combustible gas

Working of Biomass Energy Production
Step 1 — Collection of Biomass
Organic materials such as **crop waste, wood, or animal manure** are collected.

Step 2 — Processing of Biomass
The biomass is processed through **combustion, fermentation, or digestion**.

Step 3 — Energy Conversion
The stored chemical energy in biomass is converted into **heat or gas fuel**.

Step 4 — Electricity Generation
Heat or biogas is used to run **turbines or generators**, producing electricity.

Biomass Energy Systems
Examples include:

- biomass power plants
- biogas plants
- biofuel production systems

Advantages of Biomass Energy

Advantage	Explanation

Renewable	Biomass resources can be regenerated
Waste management	Uses agricultural and organic waste
Reduces landfill waste	Converts waste into useful energy
Lower carbon emissions	Emits less carbon compared to fossil fuels

Limitations of Biomass Energy

Limitation	Explanation
Air pollution	Burning biomass releases smoke and gases
Land use	Large biomass production requires land
Lower efficiency	Less efficient compared to fossil fuels

Applications of Biomass Energy

Biomass energy is used for:

- electricity generation
- biogas production for cooking
- biofuel production (ethanol, biodiesel)
- heating systems

Biomass Energy — Sources, Advantages, Limitations, Applications

Sources of Biomass Energy

Source	Explanation
Agricultural waste	Crop residues such as rice husk, straw, and corn stalks used as fuel.
Animal waste	Animal manure used to produce **biogas** through anaerobic digestion.
Wood	Wood and forest residues used for heating and electricity generation.

Advantages

Advantage	Explanation
Renewable	Biomass materials can be replenished through natural processes.
Waste utilization	Converts agricultural and organic waste into useful energy.

Limitations

Limitation	Explanation
Air pollution if poorly managed	Burning biomass inefficiently can release smoke and harmful gases.

Applications

Application	Description
Biogas plants	Convert animal waste and organic materials into biogas for cooking and electricity.
Biomass power plants	Generate electricity by burning biomass fuels such as wood and agricultural residues.

Geothermal Energy

Definition : Geothermal energy is the energy obtained from the **heat stored inside the Earth**.

The word **geothermal** comes from:

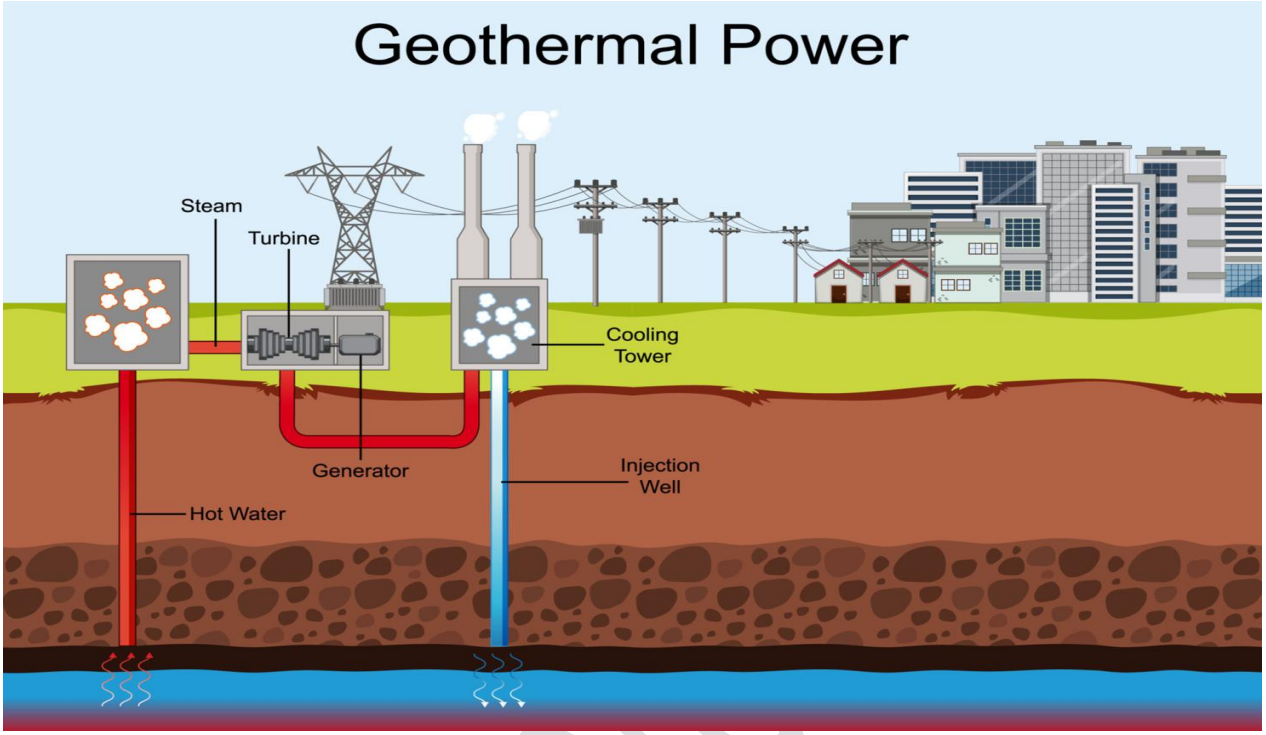
- **Geo** = Earth
- **Thermal** = Heat

This heat from the Earth's interior can be used to generate **electricity or provide heating**.

Geothermal energy is considered a **renewable and sustainable energy source** because the Earth's internal heat is continuously produced.

Principle of Geothermal Energy

Geothermal power plants work on the principle of **using heat from the Earth's interior to produce**



steam, which drives turbines to generate electricity.

The process involves converting **thermal energy from underground reservoirs into mechanical energy and then electrical energy**.

Source of Geothermal Heat

The heat inside the Earth comes from:

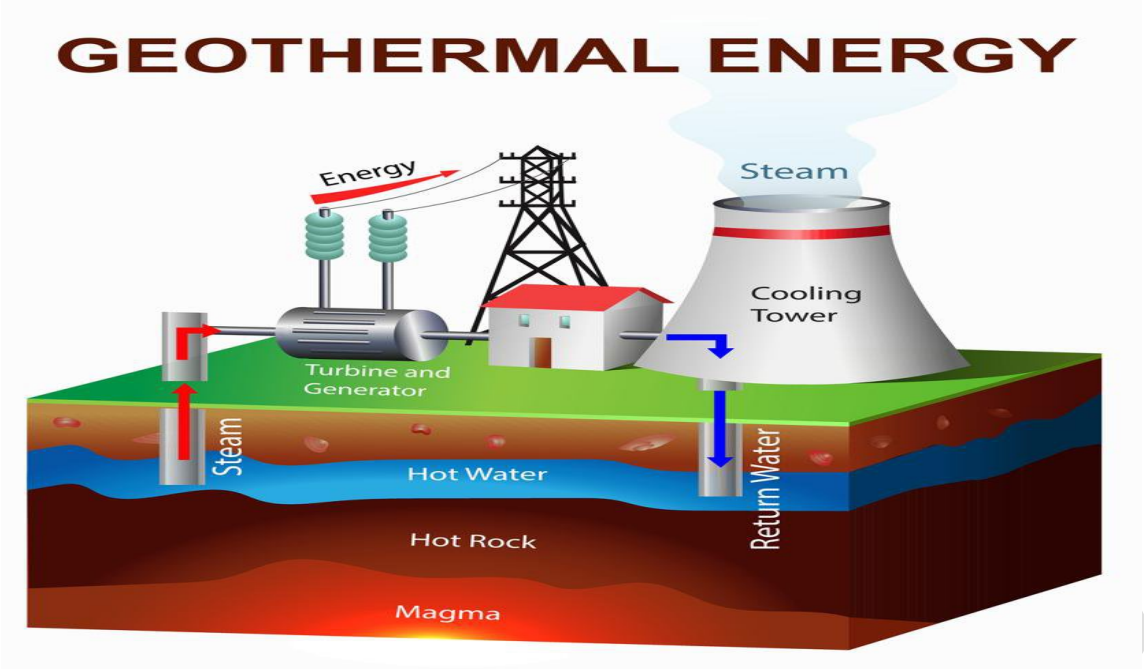
- radioactive decay of minerals
- heat remaining from Earth's formation
- magma movement beneath the Earth's crust

This heat warms underground water and forms **hot water reservoirs and steam**.

Geothermal Power Plant

Main Components of a Geothermal Power Plant

Component	Function
Geothermal reservoir	Underground source of hot water or steam
Production well	Brings hot water or steam to the surface
Turbine	Rotated by steam to produce mechanical energy
Generator	Converts mechanical energy into electricity
Injection well	Returns cooled water back underground



Working of Geothermal Power Plant

Step 1 — Heat from Earth's Interior

Heat from underground rocks warms water stored beneath the Earth's surface.

Step 2 — Steam Production

The heated water forms **high-pressure steam**.

Step 3 — Steam Movement

Steam rises through **production wells** to the surface.

Step 4 — Turbine Rotation

Steam rotates the **turbine blades**.

Step 5 — Electricity Generation

The rotating turbine drives a **generator**, producing electricity.

Step 6 — Water Re-injection

After steam cools, water is pumped back underground through **injection wells**.
This maintains the geothermal reservoir.

Applications of Geothermal Energy

Geothermal energy is used for:

- electricity generation
- heating buildings
- greenhouse heating
- industrial processes
- hot water supply

Advantages of Geothermal Energy

Advantage	Explanation
Renewable	Earth continuously produces heat
Low pollution	Produces very low greenhouse gases
Reliable	Works continuously unlike solar or wind

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Small land requirement	Requires less land compared to other power plants
Continuous energy supply	Geothermal energy is available continuously because heat from the Earth's interior is constantly produced.

Limitations of Geothermal Energy

Limitation	Explanation
Location specific	Only available in geothermal regions
High initial cost	Drilling wells is expensive
Possible gas emissions	Small amounts of gases may be released

Limitations

Limitation	Explanation
Location dependent	Geothermal power plants can only be built in areas where geothermal heat sources are available, such as volcanic or tectonically active regions.

Applications

Application	Description
Geothermal power plants	Facilities that use heat from underground reservoirs to produce steam and generate electricity.

Tidal Energy

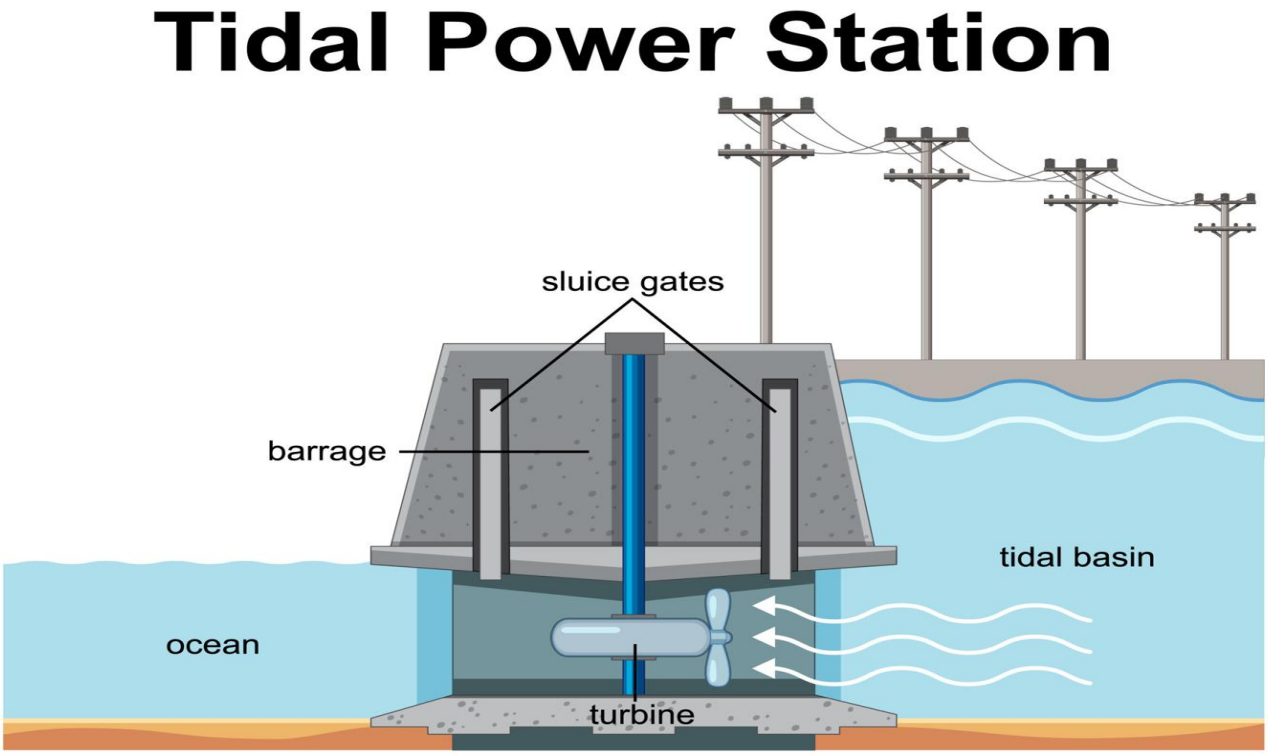
Definition : Tidal energy is the energy obtained from the **rise and fall of ocean tides**, which is used to generate electricity.

Tides occur due to the **gravitational attraction of the Moon and the Sun on Earth's oceans**.

Tidal energy is a **renewable and predictable source of energy** because tides occur regularly.

Principle of Tidal Energy

Tidal power plants work on the principle of **converting the kinetic and potential energy of moving seawater during tides into electrical energy**.



During high and low tides, large volumes of water move in and out of coastal areas. This moving water rotates **turbines**, which drive **generators to produce electricity**.

Tidal Power Plant:
Main Components of a Tidal Power Plant

Component	Function
Tidal barrage	Dam built across an estuary or coastal area
Turbine	Rotated by moving tidal water
Generator	Converts mechanical energy into electricity
Sluice gates	Control water flow during tides
Powerhouse	Contains turbine and generator

Working of Tidal Energy System
Step 1 — High Tide
During high tide, seawater enters the basin behind the **tidal barrage**.

Step 2 — Water Storage
Water is temporarily stored in the **tidal basin**.

Step 3 — Water Flow
When the tide recedes, stored water flows back toward the sea through **turbines**.

Step 4 — Turbine Rotation
The flowing water rotates the **turbine blades**.

Step 5 — Electricity Generation
The turbine drives a **generator**, producing electricity.

Types of Tidal Energy Systems

Type	Explanation
Tidal barrage	Dam across tidal estuary
Tidal stream generator	Underwater turbines in fast tidal currents
Tidal lagoon	Artificial lagoon capturing tidal energy

Advantages of Tidal Energy

Advantage	Explanation
Renewable	Tides occur naturally and continuously
Predictable	Tidal cycles are highly predictable
Clean energy	Produces very little pollution
Predictable energy source	Tides occur regularly due to gravitational forces of the Moon and Sun, making tidal energy highly predictable.

Limitations of Tidal Energy

Limitation	Explanation
High construction cost	Building tidal barrages is expensive
Location dependent	Suitable only in coastal areas with large tidal ranges
Environmental impact	May affect marine ecosystems

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High infrastructure cost	Construction of tidal barrages and underwater turbines requires large investment and complex infrastructure.

Applications of Tidal Energy

Tidal energy is used for:

- electricity generation in coastal regions
- tidal power stations
- small-scale coastal energy systems

Green Hydrogen

Definition : Green hydrogen is hydrogen fuel produced using **renewable energy sources such as solar, wind, or hydropower to split water into hydrogen and oxygen through electrolysis**.
It is called **green** because the process does **not produce carbon emissions**.
Green hydrogen is considered a **clean and sustainable energy carrier**.

Meaning of Hydrogen as Energy

Hydrogen is the **most abundant element in the universe** and can be used as a **fuel**.
When hydrogen is used as fuel, it produces:

$$H_2 + O_2 \rightarrow H_2O + Energy$$

The only by-product is **water**, making it environmentally friendly.

Principle of Green Hydrogen Production

Green hydrogen is produced using the **electrolysis of water powered by renewable energy**.
Electrolysis splits water into hydrogen and oxygen using electricity.
Steps of the principle:

1. Renewable energy (solar or wind) produces electricity.
2. Electricity is supplied to an **electrolyzer**.
3. Water molecules split into **hydrogen and oxygen**.
4. Hydrogen gas is collected and stored as fuel.

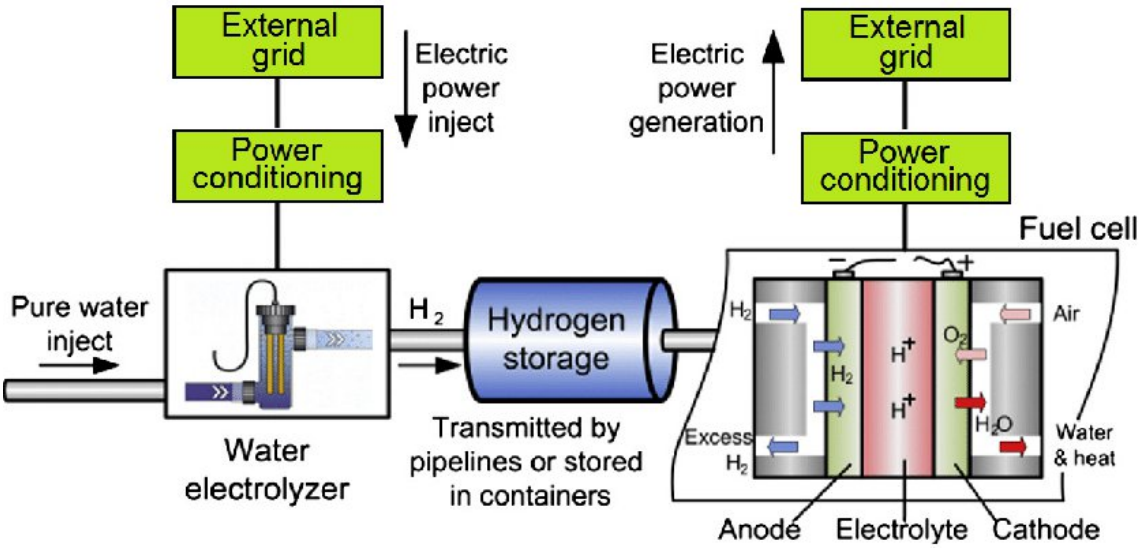
Electrolysis Process

Chemical reaction:

$$2H_2O \rightarrow 2H_2 + O_2$$

Meaning:
Water → Hydrogen + Oxygen.

Main Components of Green Hydrogen System



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rogen Production

Step 1 — Renewable Electricity
Electricity is generated using **solar panels or wind turbines**.

Step 2 — Electrolysis
Electric current passes through water in the **electrolyzer**, splitting water molecules.

Step 3 — Hydrogen Production
Hydrogen gas is produced at the **cathode**, and oxygen is produced at the **anode**.

Step 4 — Storage
Hydrogen gas is collected and stored in **high-pressure tanks**.

Step 5 — Energy Use
Hydrogen can be used in **fuel cells, vehicles, and power generation systems**.

Applications of Green Hydrogen
Green hydrogen is used in:

- hydrogen fuel cell vehicles
- electricity generation
- industrial processes (steel production)
- energy storage systems
- fertilizer production

Advantages of Green Hydrogen

Advantage	Explanation
Clean fuel	Produces only water as by-product
Renewable	Produced using renewable energy
Energy storage	Can store excess renewable energy
Versatile	Used in transport, industry, and power generation

Limitations of Green Hydrogen

Limitation	Explanation
High production cost	Electrolyzers and renewable systems are expensive
Energy intensive	Requires large amount of electricity
Storage challenges	Hydrogen storage and transport are difficult

Importance of Green Hydrogen
Green hydrogen is important for:

- reducing carbon emissions
- replacing fossil fuels in industries
- supporting clean energy transition

Many countries are investing heavily in **hydrogen energy technologies**.

Why It Is Called "Green"
Hydrogen can be produced in different ways.

Type	Production Method	Environmental Impact
Green hydrogen	Renewable energy electrolysis	No carbon emissions
Grey hydrogen	Natural gas reforming	Produces CO_2
Blue hydrogen	Fossil fuels with carbon capture	Reduced emissions

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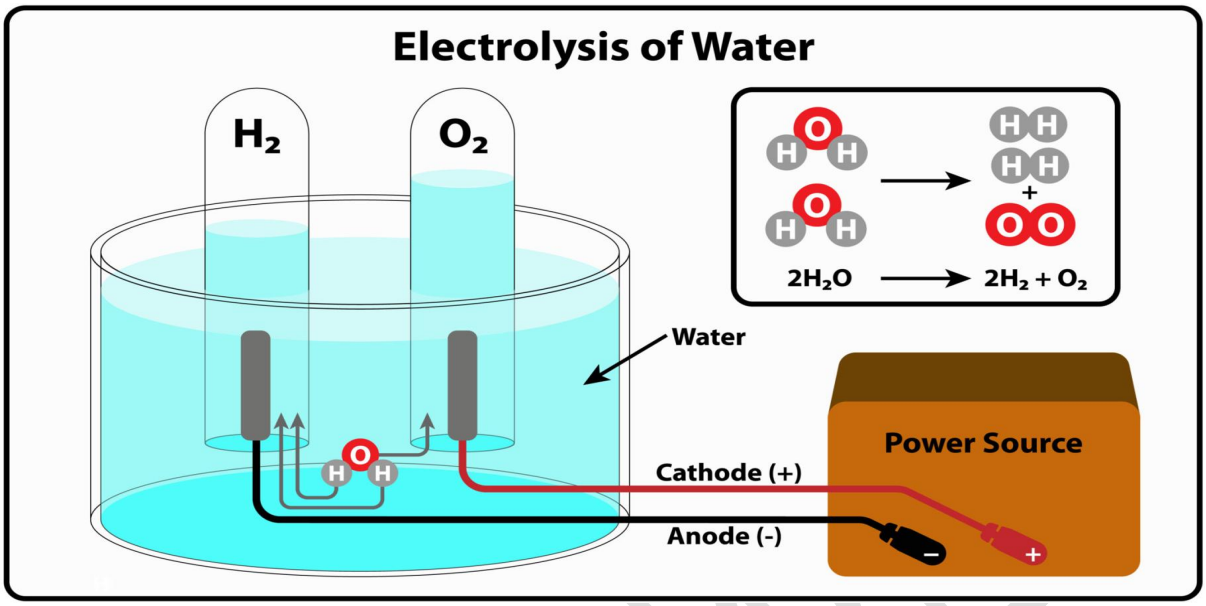
Green hydrogen is the **cleanest form of hydrogen energy**.

Importance of Green Hydrogen



Green hydrogen helps in:

- reducing greenhouse gas emissions
- replacing fossil fuels in industries
- storing renewable energy
- supporting clean transportation

Key Advantages

Advantage	Explanation
Clean fuel	Produces only water when used
Renewable	Produced using renewable energy
Energy storage	Stores excess renewable electricity

Production of Hydrogen (Electrolysis of Water Using Renewable Electricity)

Definition : Hydrogen can be produced by **electrolysis of water**, a process in which electricity is used to split water molecules into hydrogen and oxygen gases.

When the electricity used for electrolysis comes from **renewable energy sources such as solar, wind, or hydropower**, the hydrogen produced is called **green hydrogen**.

Principle of Hydrogen Production

The production of hydrogen through electrolysis is based on the principle that **electric current passing through water can break the chemical bonds between hydrogen and oxygen atoms**.

Water contains:

$$H_2O$$

When electricity flows through water, it splits into:

$$2H_2O \rightarrow 2H_2 + O_2$$

This means:

Water → Hydrogen gas + Oxygen gas.

Electrolysis Process

Electrolysis takes place in a device called an **electrolyzer**.

Main Components of Electrolysis System

Component	Function
Electrolyzer	Device where water splitting occurs
Electrodes	Conduct electricity in water
Electrolyte	Conducts electric current
Renewable electricity source	Provides electrical energy
Hydrogen storage tank	Stores produced hydrogen

Working of Electrolysis for Hydrogen Production

Step 1 — Renewable Electricity Supply

Electricity is generated from **renewable sources such as solar panels or wind turbines**.

Step 2 — Water Supply to Electrolyzer

Water is supplied to the **electrolysis chamber**.

Step 3 — Electric Current Flow

Electric current passes through the water via **two electrodes (anode and cathode)**.

Step 4 — Water Splitting

Water molecules break into hydrogen and oxygen gases.

- Hydrogen forms at the **cathode**
- Oxygen forms at the **anode**

Step 5 — Gas Collection

The hydrogen gas is **collected and stored**, while oxygen may be released or used for industrial purposes.

Chemical Reactions in Electrolysis

At Cathode:

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

At Anode:

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$$

Overall reaction:

$$2H_2O \rightarrow 2H_2 + O_2$$

Applications of Hydrogen Produced

Hydrogen produced by electrolysis can be used for:

- hydrogen fuel cell vehicles
- electricity generation
- industrial processes
- ammonia production
- energy storage systems

Advantages of Electrolysis Using Renewable Energy

Advantage	Explanation
Clean process	Produces no carbon emissions
Renewable	Uses solar or wind electricity
Energy storage	Stores excess renewable energy as hydrogen

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Limitations

Limitation	Explanation
High cost	Electrolyzers and renewable systems are expensive
Energy intensive	Requires large amount of electricity

Hydrogen Storage

Hydrogen storage is an important part of the **hydrogen energy system** because hydrogen must be stored safely before it is transported or used as fuel.
Hydrogen is a **very light gas**, so it requires special storage methods.

Definition : Hydrogen storage is the process of **storing hydrogen gas safely and efficiently for later use as a fuel or energy source**.
Hydrogen can be stored in different ways depending on the **pressure, temperature, and application**.

Methods of Hydrogen Storage

Two common storage methods are:

1. **High-pressure storage (compressed hydrogen)**
2. **Cryogenic storage (liquid hydrogen)**

1. High-Pressure Hydrogen Storage

Definition : High-pressure storage is the method of storing hydrogen gas in **strong tanks at very high pressure**.
Hydrogen gas is compressed and stored in **cylinders or tanks**.

Working Principle

Since hydrogen has **very low density**, compressing it increases the amount of hydrogen stored in a container.
Typical storage pressure:

- **350 bar**
- **700 bar**

These tanks are made of **high-strength steel or carbon fiber materials**.

Applications

High-pressure hydrogen storage is commonly used in:

- hydrogen fuel cell vehicles
- industrial hydrogen supply
- portable hydrogen storage systems

Advantages

Advantage	Explanation
Simple technology	Easier to implement compared to other storage methods
Fast filling	Tanks can be filled quickly
Suitable for vehicles	Used in hydrogen fuel cell cars

Limitations

Limitation	Explanation
High pressure required	Needs very strong tanks
Safety concerns	Risk of leakage if not handled properly

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2 Cryogenic Hydrogen Storage

Definition

Cryogenic storage is the method of storing hydrogen in **liquid form at extremely low temperatures**.
Hydrogen gas is cooled to **very low temperatures** until it becomes liquid.
Temperature required:

$$-253^{\circ}C$$

Working Principle

When hydrogen is cooled to very low temperatures, it changes from **gas to liquid**, which greatly reduces its volume.
This allows **more hydrogen to be stored in a smaller space**.

Applications

Cryogenic hydrogen storage is used in:

- space rockets
- aerospace systems
- large-scale hydrogen storage systems

Advantages

Advantage	Explanation
High storage density	Stores large amount of hydrogen
Efficient for large-scale storage	Suitable for industrial applications

Limitations

Limitation	Explanation
Extremely low temperature required	Requires special cooling systems
Energy intensive	Liquefaction requires large energy
Boil-off losses	Hydrogen may slowly evaporate

Comparison of Hydrogen Storage Methods

Feature	High-Pressure Storage	Cryogenic Storage
State of hydrogen	Gas	Liquid
Storage condition	High pressure	Very low temperature
Typical use	Vehicles	Aerospace and large storage
Complexity	Moderate	High

Importance of Hydrogen Storage

Hydrogen storage is essential for:

- hydrogen fuel cell vehicles
- renewable energy storage
- industrial hydrogen supply
- transportation of hydrogen fuel

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1. Transportation of Hydrogen

Definition : Hydrogen transportation refers to the process of **moving hydrogen from production sites to storage facilities or places where it will be used**.
Hydrogen can be transported in different forms depending on the **distance, quantity, and method of storage**.

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Methods of Hydrogen Transportation

1. Pipeline Transport

Hydrogen gas can be transported through **pipelines**, similar to natural gas pipelines.

Features

- pipelines carry compressed hydrogen gas
- suitable for **large-scale industrial transport**

Advantages

- continuous supply
- efficient for long distances

Limitations

- expensive pipeline infrastructure
- hydrogen leakage risk due to small molecule size

2. High-Pressure Cylinders

Hydrogen gas can be transported in **high-pressure cylinders or tanks** mounted on trucks.

Features

- hydrogen compressed at high pressure
- cylinders made of strong steel or carbon fiber

Applications

- hydrogen fuel stations
- laboratory supply

3. Liquid Hydrogen Transport

Hydrogen can be transported in **liquid form using cryogenic tanks**.

Hydrogen must be cooled to:

$$-253^\circ C$$

Advantages

- higher storage density
- suitable for large-scale transport

Limitations

- expensive cooling systems
- evaporation losses

4. Hydrogen Carriers (Ammonia or Liquid Organic Carriers)

Hydrogen can also be transported as **chemical compounds**.

Example:

- ammonia (NH_3)
- liquid organic hydrogen carriers (LOHC)

Later hydrogen can be extracted from these compounds.

5 Applications of Hydrogen

Hydrogen has many important applications in **energy, transportation, and industry**.

6 Fuel Cell Vehicles

Hydrogen is used in **fuel cells to power vehicles**.

Examples:

- hydrogen cars
- buses
- trucks

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Fuel cells convert hydrogen into electricity to run electric motors.

7 Electricity Generation
Hydrogen can be used in **fuel cells or gas turbines** to generate electricity.
This helps store excess renewable energy.

8 Industrial Applications
Hydrogen is widely used in industries such as:

- petroleum refining
- fertilizer production (ammonia synthesis)
- chemical manufacturing
- steel production

9 Energy Storage
Hydrogen can store **excess renewable energy**.
Example:
solar energy → hydrogen → electricity later.
This helps balance energy supply and demand.

10 Aerospace Applications
Hydrogen is used as a **rocket fuel** in space missions.
Example:
liquid hydrogen used in spacecraft propulsion systems.

Advantages of Hydrogen as Energy Carrier

Advantage	Explanation
Clean fuel	Produces only water when used
High energy content	Contains high energy per unit mass
Versatile	Used in transport, electricity, and industry

Limitations of Hydrogen Use

Limitation	Explanation
Storage difficulty	Hydrogen is very light and requires special storage
High infrastructure cost	Transport and storage systems are expensive

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